7.8.3 Protocol data unit

7.8.3.1 Structure

The V2GTP PDU consists of a header and a body section as shown in Table 11.

Table 11 — V2GTP message structure

Header	Payload
8 Bytes	4294967295 Bytes

The payload contains the application data (e.g. a V2G message). The header separates the payloads (i.e. individual V2GTP messages) within a byte stream and gives information for the payload processing.

The V2GTP message header structure is shown in Table 12 and described in Table 13. The supported payload types are described in Table 14.

Table 12 — V2GTP message header structure

Byte No.	1	2	3	4	5	6	7	8
Header field	Protocol version	Inverse protocol version	Payload type		Payload length			

[V2G20-082] A V2GTP entity shall use the header structure as shown in Table 12.

[V2G20-083] A V2GTP entity shall send the 8 bytes of the V2GTP header in the order as shown in Table 12.

[V2G20-084] A byte with a lower number shall be sent before a byte with a higher number. The header starts with byte 1 and ends with byte 8.

[V2G20-085] A V2GTP entity shall send the fields "payload type" and "payload length" in big-endian format: The most significant byte is sent first, the least significant byte is sent last.

Table 13 — Generic V2GTP header structure

Header field	Header field description	Header field values
Protocol Version	Identifies the protocol version of V2GTP messages.	0x01: V2GTP version 1 0x00, 0x02-0xFF: reserved by document
Inverse Protocol Version	Contains the bit-wise inverse value of the protocol version which is used in conjunction with the V2GTP protocol version as a protocol verification pattern to ensure that a correctly formatted V2GTP message is received. Equals the <Protocol_Version> XOR 0xFF	0xFE: V2GTP Version 1 0xFF, 0xFD-0x00: reserved by document
Payload Type	Contains information about how to decode the payload following the V2GTP header.	Refer to Table 14 for a complete list of payload type values.
Payload Length	Contains the length of the V2GTP message payload in bytes (i.e. excluding the generic V2GTP header bytes).	0...4294967295

Table 14 — V2GTP payload types

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Payload type value	Payload type identifier	Explanation
0x0000 - 0x8000	Not applicable	Reserved for future use
0x8001	SAPPayloadID	EXI encoded V2G messages of ISO 15118-2. For backward compatibility the supportedAppProtocol messages of this document also use this payload type (see 8.2.1)
0x8002	Part20MainstreamPayloadID	EXI encoded V2G messages of this document which are exchanged as part of the main stream of the communication flow. This set of messages is using the XSD schema with the namespace "urn:iso:std:iso:15118:-20:CommonMessages".
0x8003	Part20ACMainstreamPayloadID	EXI encoded V2G messages of this document which are exchanged as part of the main stream of the communication flow and are specific to AC. This set of messages is using the XSD schema with the namespace "urn:iso:std:iso:15118:-20:AC".
0x8004	Part20DCMainstreamPayloadID	EXI encoded V2G messages of this document which are exchanged as part of the main stream of the communication flow and are specific to DC. This set of messages is using the XSD schema with the namespace "urn:iso:std:iso:15118:-20:DC".
0x8005	Part20ACDPMainstreamPayloadID	EXI encoded V2G messages of this document which are exchanged as part of the main stream of the communication flow and are specific to ACDP. This set of messages is using the XSD schema with the namespace "urn:iso:std:iso:15118:-20:ACDP".
0x8006	Part20WPTMainstreamPayloadID	EXI encoded V2G messages of this document which are exchanged as part of the main stream of the communication flow and are specific to WPT. This set of messages is using the XSD schema with the namespace "urn:iso:std:iso:15118:-20:WPT".
0x8007 - 0x8100	Not applicable	Reserved for future use
0x8101	ScheduleRenegotiationPayloadID	EXI encoded V2G message on a multiplexed side stream for the purpose of schedule renegotiation (see 8.6.5.1 and Annex F)
0x8102	MeteringConfirmationPayloadID	EXI encoded V2G message on a multiplexed side stream for the purpose of exchanging Metering information (see 8.3.4.3.11)
0x8103	ACDPSystemStatusPayloadID	EXI encoded V2G message on a multiplexed side stream for the purpose of exchanging ACDP system status information (see 8.3.4.7.8)

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Payload type value	Payload type identifier	Explanation
0x8104	ParkingStatusPayloadID	EXI encoded V2G message on a multiplexed side stream for the purpose of delivering Parking Status information (see 8.6.5.3 and 8.3.4.8.1)
0x8105 - 0x8FFF	Not applicable	Reserved for future use
0x9000	SDPRequestPayloadID	SDP request message (see 7.10.1.5)
0x9001	SDPResponsePayloadID	SDP response message (see 7.10.1.6)
0x9002	SDPRequestWirelessPayloadID	SDP request message for wireless communication (see 7.10.1.8)
0x9003	SDPResponseWirelessPayloadID	SDP response message for wireless communication (see 7.10.1.9)
0x9004 - 0x9FFF	Not applicable	Reserved for future use
0xA000 - 0xFFFF	Not applicable	Available for manufacturer specific use. Uniqueness of those identifiers is not guaranteed by this document.

NOTE 1 Only the payload type value numbers are technically used by the protocol. However, the requirements of this document reference those values by the associated identifier with the PayloadID suffix.

[V2G20-086] A V2GTP entity shall use the V2GTP message structure as shown in Figure 8 to send V2G messages as defined in Clause 8.

[V2G20-2307] A V2GTP entity shall use the definitions as defined in Table 13 and Table 14.

[V2G20-1229] For EXI encoded V2G messages (payload 0x8001 up to and including 0x8FFF) a V2GTP entity shall use a separate V2GTP message for each V2G message.

NOTE 2 Requirement [V2G20-1229] implies that the payload field can include neither a part of a message nor multiple messages.

7.8.3.2 Header processing

For the processing of the payload the V2GTP entity processes the header first. For this, a V2GTP entity that receives a V2GTP message checks the header field step by step. The header processing as defined below is illustrated in Figure 11.

[V2G20-089] A V2GTP entity shall process the V2GTP header as defined in Table 13 before processing the payload as defined in Table 14.

[V2G20-090] A V2GTP entity shall check the protocol version and inverse protocol version fields (synchronization pattern) before any other header fields.

[V2G20-092] A V2GTP entity shall check the payload type after the successful check of the version and inverse version field.

[V2G20-094] A V2GTP entity shall check the payload length after the successful check of the payload type.

[V2G20-096] If the header processing was successful the V2GTP entity shall process the payload.

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